

Title Page

Manuscript Title:

Commissioning-Calibrated GP-HOCBF Safety Filtering for Ultra-Supercritical Boiler-Turbine Control under Model Mismatch

Article Type:

Original Research Article

Authors:

Zhuang Shao^{1,*}, Lijun Lei², Peng Wang³, Liang Zheng², Jie Zhou²

Affiliations:

1 China Resources Power Technology Research Institute Co., Ltd., Shenzhen 518000, Guangdong Province, China

2 Rundian Energy Science and Technology Co., Ltd., Zhengzhou 450052, Henan Province, China

3 North China University of Water Resources and Electric Power, Zhengzhou 450046, Henan Province, China

Corresponding Author:

Zhuang Shao

China Resources Power Technology Research Institute Co., Ltd.

Shenzhen 518000, Guangdong Province, China

Email: shaozhuang@crpower.com.cn

ORCID:

Zhuang Shao: 0000-0003-2496-0797

Acknowledgements:

Not applicable.

Funding:

This work was supported by the Postdoctoral Research Project of China Resources Power Technology Research Institute Co., Ltd. (Internal Project No. 374322).

Declaration of Conflicting Interests:

The author(s) declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

Data Availability Statement:

The source code, simulation scripts, benchmark results, plotting scripts, derived plant-historian metrics, and anonymized controller-export excerpts are available at <https://github.com/VickylastShao/Robust-HOCBF-Safety-Filtering-for-Supercritical-Power-Plants-under-Model-Mismatch>.

Complete raw plant records are proprietary enterprise assets. Qualified researchers may request restricted access from the corresponding author, subject to data-owner approval and an executed data-use agreement.

Author Contributions:

Zhuang Shao conceived the study, developed the methodology, performed the simulations, analyzed the results, and wrote the original draft.

Lijun Lei provided engineering requirements, plant-data interpretation, and field-validation support.

Peng Wang contributed to control-system analysis and manuscript review.

Liang Zheng supported plant-data preparation and validation-log interpretation.

Jie Zhou contributed to deployment-context analysis and manuscript review.

All authors reviewed and approved the final manuscript.

Number of words:

8630

Number of figures:

8

Number of tables:

5